

End-to-end differentiable inverse design of disordered structures with physics-guided diffusion model and Tesseract

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Tesseract Hackathon - Track 1: Inverse design & shape optimization

1 Problem formulation

In this project, we aim to design 2D disordered foam structures with a specified effective thermal conductivity. We consider a simple steady-state heat conduction setup here. The computational domain represents a region with dimension $1 \times 1 \text{ m}^2$ between two walls, containing a disordered foam structure and voids filled with gas, as illustrated in Fig. 1a. The top boundary is maintained at temperature $T_1 = 500 \text{ K}$, while the bottom boundary is held at $T_2 = 300 \text{ K}$. The left and right walls are assumed to be adiabatic.

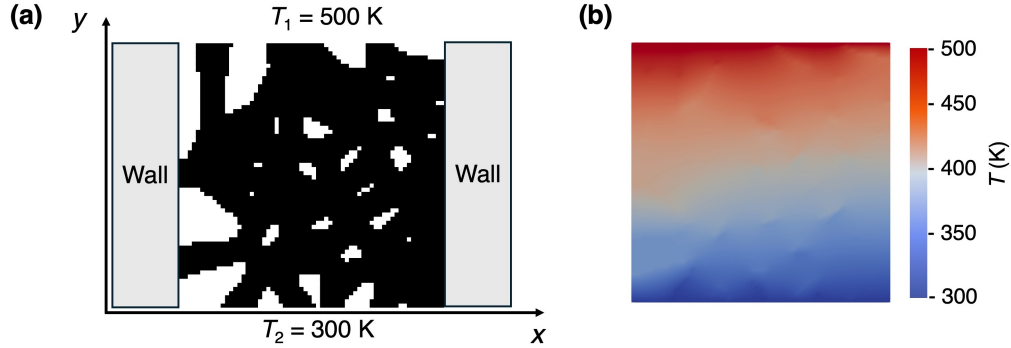


Figure 1: Problem formulation.

The governing equation for steady-state heat conduction is given by:

$$-\nabla \cdot (k \nabla T) = 0 \quad (1)$$

with k denotes the local thermal conductivity and T is the temperature field. As a simplification, we treat both materials as having constant thermal conductivities, where $100 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ for the solid phase and $1 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ for the gas phase. The boundary conditions are specified as follows:

$$\begin{aligned} T &= T_1 && \text{on } \Gamma_D^{\text{top}} \\ T &= T_2 && \text{on } \Gamma_D^{\text{bottom}} \\ -k \frac{\partial T}{\partial \mathbf{n}} &= 0 && \text{on } \Gamma_N^{\text{wall}} \end{aligned} \quad (2)$$

where $\mathbf{n}$ represents the outward unit normal vector to the boundary, Γ_D^{top} and Γ_D^{bottom} denote the top and bottom Dirichlet boundaries, Γ_N^{wall} represents the adiabatic walls.

A finite element method (FEM) solver is used here to solve for the steady-state temperature field, and an example steady-state solution is shown in Fig. 1b. Then, the effective thermal conductivity of the structure is given as [1]:

$$k_{\text{eff}} = \frac{1}{\Delta T} \int_L k \left| \frac{\partial T}{\partial y} \right|_{y=\text{const}} dx \quad (3)$$

where k_{eff} is the effective thermal conductivity, ΔT is the temperature difference, L is the side length perpendicular to heat flow, and y and x represent the directions parallel and perpendicular to the heat flow, respectively. In this work, the integration is performed over the top surface of foams.

2 Tesseract and workflow

The inverse design of disordered structures presents a formidable challenge. Unlike periodic structures where properties can be engineered through systematic modification of a few unit-cell parameters, disordered systems lack such clear design principles. The structure-property relationships are highly complex and non-intuitive, making designing disordered structures with targeted properties exceedingly difficult and rendering traditional trial-and-error approaches impractical. Herein, we utilize the power of diffusion models and differentiable programming to handle the problem.

The end-to-end inverse design framework for this project has four components: Diffusion model, Manufacturability module, Mesher and FEM solver. This pipeline is fully differentiable with the help of an open source software Tesseract, which is applied as the connector between different components. The derivatives can be defined and backpropagated from the downstream component to the upstream ones through Tesseract.

The details of these four components are introduced in the subsections below.

2.1 Tesseract 1: Diffusion model

Diffusion model receives a Gaussian noise as the input and performs the denoising/sampling process to restore the clean image. In this project, the diffusion model is trained with disordered foam structures generated with Voronoi tessellation [2, 3]. Diffusion model is responsible for providing structural prior for the inverse design task.

As the diffusion model in this project is unconditional, it can only generate random foam structures without considering the property. Therefore, we add physics-guidance [4] to guide the generation during sampling. To design disordered foam structures with a target effective thermal conductivity, the sampling stochastic differential equation (SDE) of the trained diffusion model is:

$$d\mathbf{x} = \left[-\frac{1}{2}\beta(t)\mathbf{x} - \beta(t) \left(-\frac{\epsilon_{\theta}(\mathbf{x}_t, t)}{\sigma_t} - \underbrace{s \nabla_{\mathbf{x}_t} \|k_{\text{eff}}^* - k_{\text{eff}}(\hat{\mathbf{x}}_0(\mathbf{x}_t))\|^2}_{\text{Physics guidance}} \right) \right] dt + \sqrt{\beta(t)} d\bar{\mathbf{w}} \quad (4)$$

where k_{eff}^* is the target effective thermal conductivity, and $k_{\text{eff}}(\cdot)$ is the physics-based solver used to calculate the effective thermal conductivity of the generated structures. Gradients are computed via customized reverse-mode automatic differentiation (AD), where specialized vector-Jacobian product (VJP) rules are derived from the implicit function theorem [5]. More information of the physics-guided diffusion models can be referred to Ref. [4]. This component is implemented with JAX [6].

2.2 Tesseract 2: Manufacturability module

To make the training of diffusion model lightweight, the size of the training images of foam structures is 64*64. In such a small resolution, the structures have jagged edges. Besides, the intermediate

structures generated during the sampling are noisy, which is not appropriate to input them to solvers. Therefore, we apply a series of morphology manipulations to optimize the generated raw structure, including upsampling, filtering, projection and connectivity check. Fig. 2 is an illustration of processing an intermediate structure with the manufacturability module. This component is built upon SciPy and scikit-image.

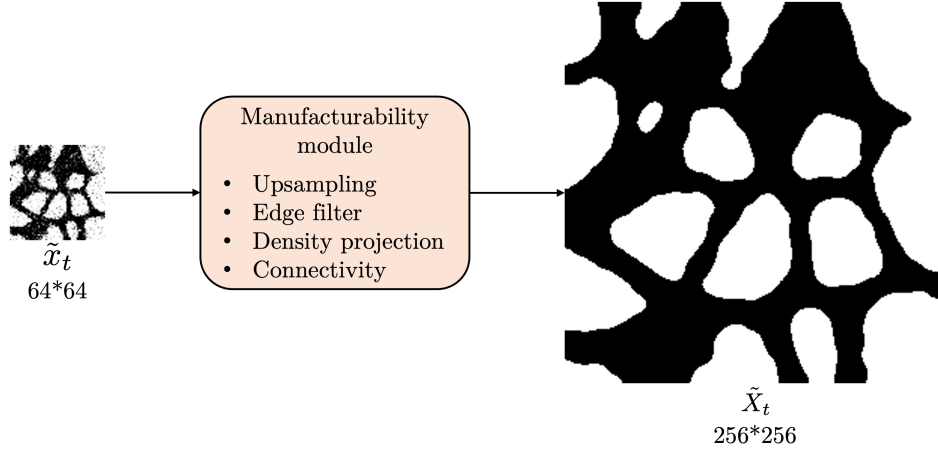


Figure 2: The manufacturability module.

2.3 Tesseract 3: Mesher

Before solving the governing equation on the generated structures, we need to mesh it. When the image size is 64×64 , we take each pixel as a natural quadrilateral mesh grid. When the image is projected by the manufacturability module, we first calculate the signed distance field to get the continuous field and find the contours. We then use an advanced mesher to better mesh the structure with triangular elements. The aim is to make the grid points fit the continuous curves and reduce the discretization error. Fig. 3 is an illustration of quadrilateral and triangular meshes on the structures. This component is implemented with Gmsh.

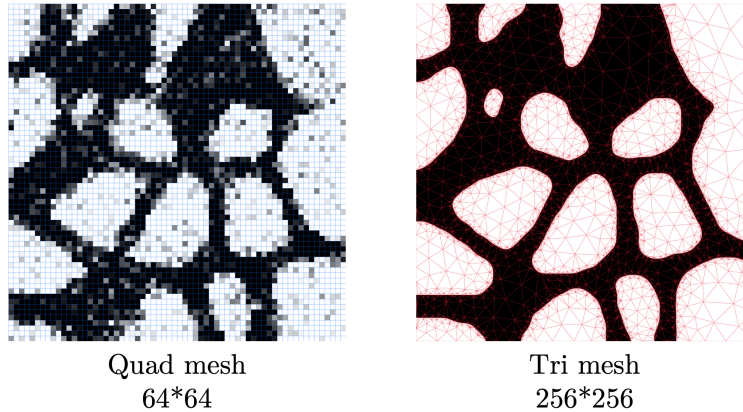


Figure 3: Meshes on the structures.

2.4 Tesseract 4: FEM solver

The last component is the numerical solver based on FEM. Here, the mesh structure and its projected property are input to the solver to calculate the temperature field and the effective thermal conductivity. The L_2 loss between the calculated and the target effective thermal conductivity is

the objective value, whose gradient is the physics-guidance term in Eq. (4). This component is implemented with JAX.

2.5 Workflow

The overall workflow of this project is shown in Fig. 4. There are 50 sampling steps for generating a structure with desired property, and these steps are further divided into two stages:

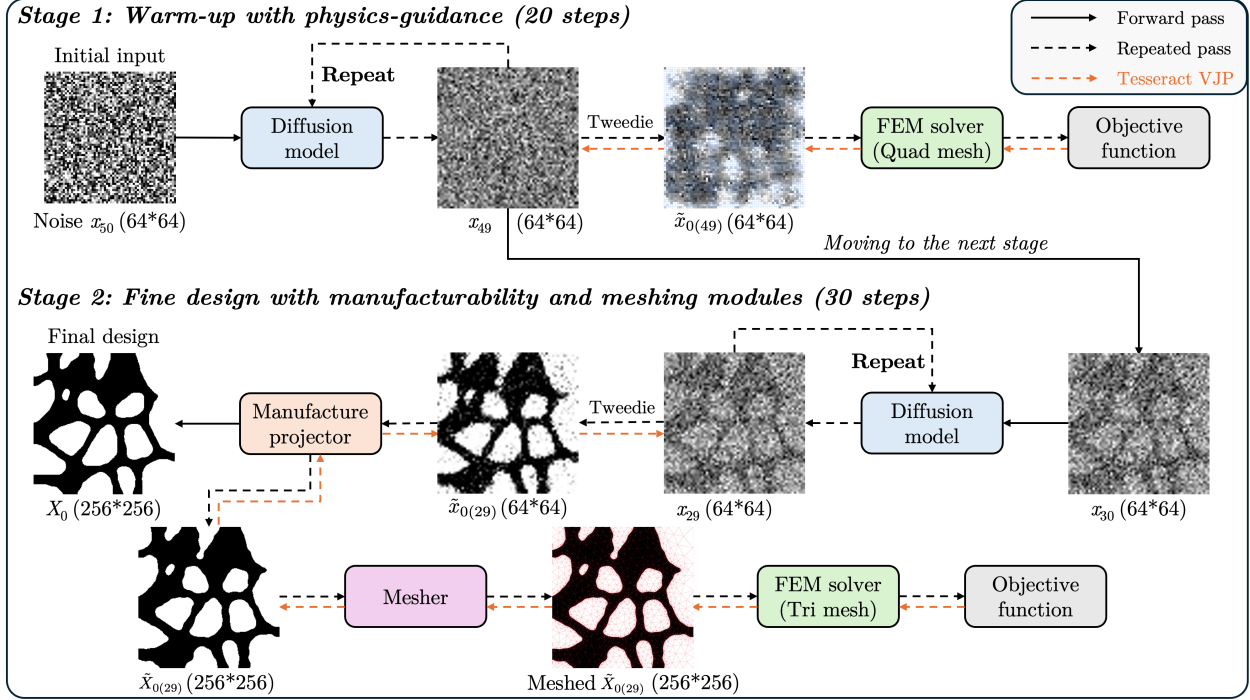


Figure 4: Framework workflow.

The first stage is the warm-up stage, which covers the former 20 steps of sampling. The task of this stage is to find a rough design which may have the trend to have the specified effective thermal conductivity. In this stage, only the diffusion model and the FEM solver are working. This stage starts with a random noise x_{50} with resolution of 64*64 feeding into the diffusion model. In the physics-guided sampling loop, the diffusion model predicts the denoised image, and then this denoised image is transformed into the clean expectation of the final output $\tilde{x}_0$. For example, in step 1, the predicted image based on input x_{50} is x_{49} , the expectation of the clean version of x_{49} is denoted as $\tilde{x}_{0(49)}$. Then a FEM solver is used to solve for the temperature field and effective thermal conductivity based on the pixel (quadrilateral) mesh. The objective is defined as the L_2 loss between the designed and targeted effective thermal conductivity. The gradient of the objective will be backpropagated through the FEM solver to the diffusion model as the physics-based guidance to modify the denoised prediction.

The second stage is the fine design stage, which covers the later 30 steps of sampling. The task of this stage is to design the structure in detail and make sure it has the specified effective thermal conductivity. The output of the first stage x_{30} will be the input here. Aside of the diffusion model and the FEM solver, a manufacturability projector and a mesher are added. The manufacturability projector upsamples and smoothes the predicted structure to make sure it is processable, and the mesher meshes the structure with finer triangular meshes to reduce the discretization error. For example, in step 21, x_{30} is input into the diffusion model, and the output and clean expectation are denoted as x_{29} and $\tilde{x}_{0(29)}$. After the manufacturability module projection, the structure, denoted

as $\tilde{X}_{0(29)}$, has a resolution of 256*256 with smoother edges. Then the mesher meshes it and the FEM solver solves the temperature field and effective thermal conductivity. The gradient of the objective will be backpropagated through the FEM solver, mesher and manufacturability projector to the diffusion model as the physics-based guidance to modify the denoised prediction.

To guarantee this pipeline is fully differentiable, we have made some approximations on the VJP rules as some manipulations are not continuous: (1) In the manufacturability module, the connectivity check may delete some disconnected pixels, so the gradient of it is not taken into account; (2) In the backward pass of contouring, we use analytical marching squares to calculate the gradient; (3) In the Mesher, the number and connection of the mesh elements are discrete, so we assume they are fixed and only the gradient of loss with respect to the point coordinates is considered.

3 Why Tesseract

This project requires Tesseract because the complete inverse-design loop combines several heterogeneous libraries and execution models that cannot be treated as one native differentiable program. The diffusion model and the FEM solver are implemented with JAX and use automatic differentiation through the SDE update, finite-element assembly, linear solve, material conductivity, and mesh coordinates. Manufacturing processing uses NumPy, SciPy.ndimage, and scikit-image for filtering, connected-component analysis, morphology, upsampling, contour extraction, and signed-distance construction; these operations include discrete forward constraints and an explicitly defined surrogate VJP for the backward pass. Gmsh is an external, system-level meshing engine. Its mesh topology is therefore treated as frozen during each reverse pass, while an explicit mesh-motion VJP propagates FEM coordinate sensitivities through the Gmsh interface, contour representation, and manufacturing field. Tesseract connects these tools through schema-defined component interfaces and VJP endpoints: Tes1-diffusion performs the guided sampling update, Tes2-manufacture returns the manufacturable level set and its surrogate gradient, Tes3-mesher returns the mesh and the coordinate-to-shape VJP, and Tes4-fem returns the physical objective together with material and coordinate sensitivities. Thus, Tesseract provides the contract and orchestration layer that makes automatic differentiation, surrogate sensitivities, analytical derivatives, and heterogeneous libraries composable within one reproducible gradient-driven design workflow.

4 Numerical result: Design of effective thermal conductivity

We present the inverse design task of 2D disordered foam structure with a target effective thermal conductivity of $30 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$. The design process are shown in Fig. 5. We can see at the warm-up stages, the generated structures is vague, and undergoing gradual modifications under the physics-guidance. At the fine design stage, the manufacturability projector makes the structure clean and smooth, enable mesher to draw finer meshes on it.

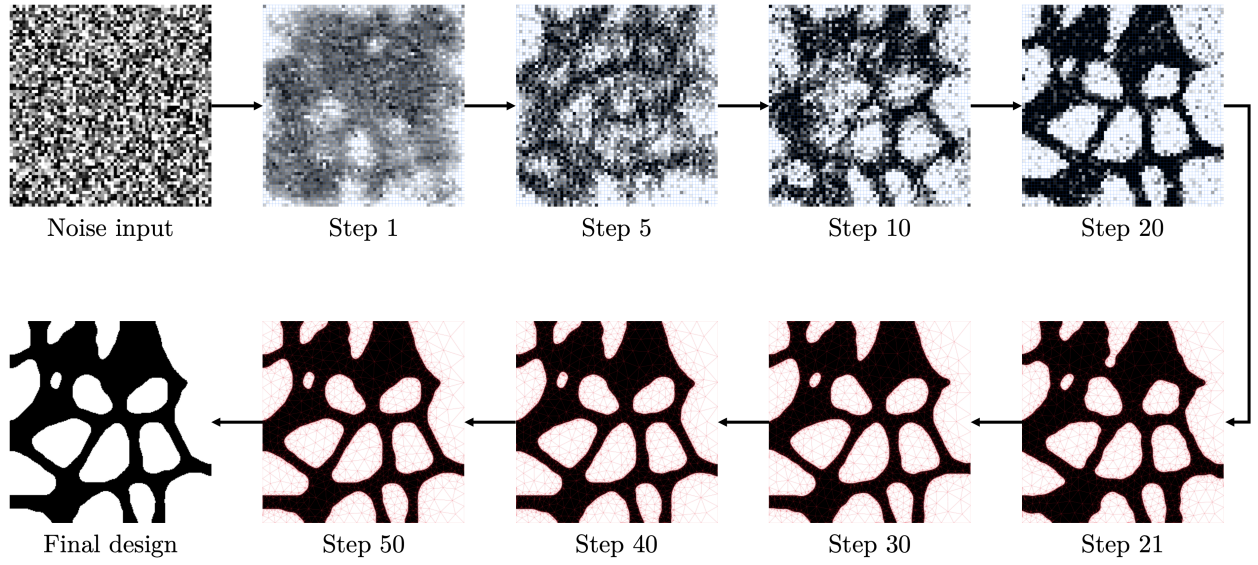


Figure 5: Design flow.

The convergence history of effective thermal conductivity toward a target effective thermal conductivity is provided in Fig. 6. The sampled foam initially shows significant conductivity fluctuations. As sampling proceeds, dominant heat-transfer paths emerge, and the structures begin to undergo only minor adjustments. Eventually, the effective thermal conductivity stabilize around the target value.

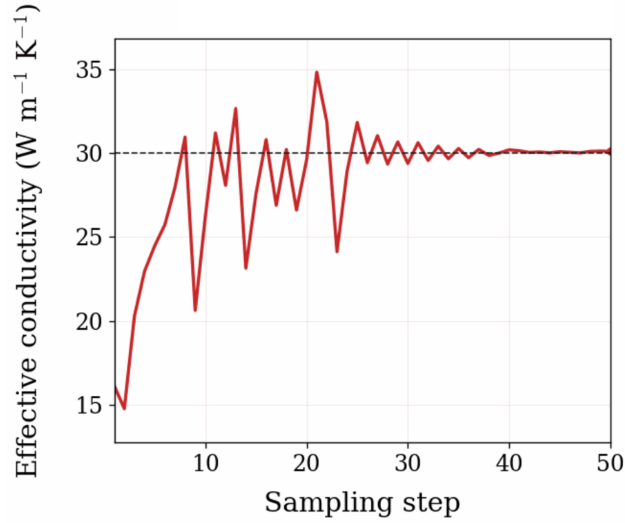


Figure 6: Flow of designed effective thermal conductivity.

Code and data availability

The underlying code and data for this study is available at <https://github.com/xiezy964/tes-phy-guide>.

References

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